

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY HYDERABAD
B.TECH. MINOR IN DATA SCIENCE

COURSE STRUCTURE AND SYLLABUS

(Applicable for the batches admitted from the academic year 2022-2023)

V SEMESTER

R22

Course Code	Title of the Course	L	T	P/D	CH	C
22MC1DS301	Fundamentals of Data Science	3	0	0	3	3
22MC2DS301	Python Programming Laboratory	0	0	3	3	1.5
Total		3	0	3	6	4.5

VI SEMESTER

R22

Course Code	Title of the Course	L	T	P/D	CH	C
22MC1DS302	Data Science Models and Applications	3	1	0	4	4
Total		3	1	0	4	4

VII SEMESTER

R22

Course Code	Title of the Course	L	T	P/D	CH	C
22MC1DS401	Data Wrangling and Visualization Strategies	3	0	0	3	3
22MC2DS401	Data Wrangling and Visualization Strategies Laboratory	0	0	3	3	1.5
Total		3	0	3	6	4.5

VIII SEMESTER

R22

Course Code	Title of the Course	L	T	P/D	CH	C
22MC1DS402	Time Series Analysis and Forecasting	3	0	0	3	3
22MP1DS401	Mini – Project	0	0	4	4	2
Total		3	0	4	7	5

L – Lecture T – Tutorial P – Practical D – Drawing
 C – Credits SE – Sessional Examination CA – Class Assessment
 SEE – Semester End Examination D-D – Day to Day Evaluation
 CP – Course Project PE – Practical Examination

CH – Contact Hours/Week
 ELA – Experiential Learning Assessment
 LR – Lab Record

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

B.Tech. Minor (DS) V Semester

(22MC1DS301) FUNDAMENTALS TO DATA SCIENCE

TEACHING SCHEME		
L	T/P	C
3	0	3

EVALUATION SCHEME				
SE	CA	ELA	SEE	TOTAL
30	5	5	60	100

COURSE OBJECTIVES:

- To learn concepts, techniques, and tools they need to deal with various facets of data science practice, including data collection and integration
- To understand the basic types of data and basic statistics
- To exploring data analysis, predictive modeling, descriptive modeling, data product creation, evaluation, and effective communication
- To identify the importance of data reduction and data visualization techniques

COURSE OUTCOMES: After completion of the course, the student should be able to
CO-1: Understand basic terms, what Statistical Inference means, Identify probability distributions commonly used as foundations for statistical modeling and fit a model to data

CO-2: Describe the data using various statistical measures

CO-3: Utilize R elements for data handling

CO-4: Perform data reduction and apply visualization techniques

COURSE ARTICULATION MATRIX:

(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial)

UNIT-I:

What is Data Science? Three pillars of data science, Types of Data, Cumulative Distribution Function, Normal Distribution, Standard Normal Distribution, Empirical Rule, and Related Problems, Assessing Normality, Binomial Distribution, Poisson Distribution, Uniform distribution, Exponential distribution, lognormal distribution

UNIT-II:

Central limit theorem, K-S Test for similarity of two distributions, power law and pareto distribution, box-cox transform, Interpretation of Chebyshev's inequality, Descriptive statistics, Inference statistics, Measures of Central Tendency, kurtosis, skewness.

UNIT-III:

Classification: Conditional probability, example of condition probability, independent events, mutually exclusive events, Bayes theorem and related problems, Naive Bayes algorithm and its mathematical interpretation, problem on Naive Bayes algorithm. Feature importance and interpretability. Imbalanced data, outliers in Naive Bayes, multiclass classification.

UNIT-IV:

Regression, and its basics. Types of regression, regression process. Linear Regression: Empirical model, Geometrical interpretation of linear regression, derivation, Logistic Regression.

UNIT-V:

Why learn PCA, Geometric intuition of PCA, Eigen values and Eigen vectors, visualizing MNIST dataset, Limitations of PCA, PCA code example, PCA for dimensionality reduction.

TEXT BOOKS:

1. Doing Data Science, Straight Talk from The Frontline, Cathy O'Neil and Rachel Schutt, O'Reilly, 2014
2. Data Mining: Concepts and Techniques, Jiawei Han, Micheline Kamber and Jian Pei, 3rd Edition, The Morgan Kaufmann Series in Data Management Systems
3. Statistical programming in R, K. G. Srinivas, G. M. Siddesh, Oxford Publications

REFERENCES:

1. Introduction to Data Mining, Pang-Ning Tan, Vipin Kumar, Michael Steinbach, Pearson Education
2. A Handbook of Statistical Analysis Using R, Brian S. Everitt, 2nd Edition, 4 LLC, 2014
3. Introductory statistics with R, Dalgaard Peter, Springer Science & Business Media, 2008
4. R Cookbook, Paul Teetor, O'Reilly, 2011

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

B.Tech. Minor (DS) V Semester

(22MC2DS301) PYTHON PROGRAMMING LABORATORY

TEACHING SCHEME		
L	T/P	C
0	3	1.5

EVALUATION SCHEME					
D-D	PE	LR	CP	SEE	TOTAL
10	10	10	10	60	100

COURSE OBJECTIVES:

- To introduce visual perception and core skills for visual analysis
- To develop skills to both design and critique visualizations
- To obtain a comprehensive knowledge of various tools and techniques for data science
- To analyses the various methods of data collection

COURSE OUTCOMES: After completion of the course, the student should be able to

CO-1: Understand the basic concepts of NumPy and Pandas

CO-2: Apply and interpret fundamental concepts of data science

CO-3: Explore the basics of data visualization

CO-4: Implementation of advanced data visualization

COURSE ARTICULATION MATRIX:

(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial)

LIST OF EXPERIMENTS / PROGRAMS:

WEEK 1,2:

1. Create NumPy arrays from Python Data Structures, Intrinsic NumPy objects and Random Functions.
2. Manipulation of NumPy arrays- Indexing, Slicing, Reshaping, Joining and Splitting.
3. Computation on NumPy arrays using Universal Functions and Mathematical methods.
4. Import a CSV file and perform various Statistical and Comparison operations on rows/columns.
5. Load an image file and do crop and flip operation using NumPy Indexing.

WEEK 3, 4, 5:

1. Create Pandas Series and Data Frame from various inputs.
2. Import any CSV file to Pandas Data Frame and perform the following:
 - a) Visualize the first and last 10 records.
 - b) Get the shape, index and column details.
 - c) Select/Delete the records (rows)/columns based on conditions.
 - d) Perform ranking and sorting operations.
 - e) Do required statistical operations on the given columns.
 - f) Find the count and uniqueness of the given categorical values.
 - g) Rename single/multiple columns.

WEEK 6, 7, 8:

1. Develop a model on residual analysis of simple linear regression.
2. Residual plots of linear regression

3. Normal probability plots.
4. Empirical model of linear regression analysis.

WEEK 9, 10, 11:

1. Import any CSV file to Pandas Data Frame and perform the following:
 - a) Handle missing data by detecting and dropping/ filling missing values.
 - b) Transform data using apply () and map() method.
 - c) Detect and filter outliers.
 - d) Perform Vectorized String operations on Pandas Series.
2. Implement regularized Linear regression.
- e) Develop a model on logistic regression on any data set for prediction.

WEEK 12, 13 & 14:

1. Visualize data using Line Plots, Bar Plots, Histograms, Density Plots and Scatter Plots.
2. Download the House Pricing dataset from Kaggle and map the values to 23 Aesthetics.
3. Use different Color scales on the Rainfall Prediction dataset.
4. Create different bar plots for variables in any dataset.
5. Show an example of Skewed data and removal of skewedness.
6. For a sales dataset do Time Series visualization.
7. Build a Scatterplot and suggest dimension reduction.

WEEK 15: LAB INTERNAL.**TEXTBOOKS:**

1. Computational and Inferential Thinking: The Foundations of Data Science, Adi Adhikari and John De Nero, 1st Edition, 2019
2. Doing Data Science, Straight Talk from The Frontline, Cathy O'Neil and Rachel Schutt, O'Reilly, 2014
3. Fundamentals of Data Visualization: A Primer on Making Informative and Compelling Figures, Claus Wilke, 1st Edition, O'Reilly, 2019

REFERENCES:

1. Mining of Massive Datasets, v2.1, Jure Leskovek, Anand Rajaraman and Jeffrey Ullman, Cambridge University Press, 2014
2. Practical Statistics for Data Scientists: 50 Essential Concepts, Bruce Peter, and Andrew Bruce, O'Reilly, 2017
3. Data Analysis and Visualization Using Python: Analyze Data to Create Visualizations for BI Systems, Ossama Embarak, Press, 2018

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

B.Tech. Minor (DS) VI Semester

(22MC1DS302) DATA SCIENCE MODELS AND APPLICATIONS

TEACHING SCHEME		
L	T/P	C
3	1	4

EVALUATION SCHEME				
SE	CA	ELA	SEE	TOTAL
30	5	5	60	100

COURSE OBJECTIVE:

- To provide deep knowledge of Data Science and how it can be applied in various fields to make the life easy
- To obtain a comprehensive knowledge of various applications of Data Science
- To understand the concept of PCA and its related applications
- To analyze the use of data in various Data Science applications i.e., in education, social media, health care, bioinformatics

COURSE OUTCOMES: After completion of the course, the student should be able to

CO-1: Correlate the data science and solutions to modern problem

CO-2: Decide when to use which type of technique in data science

CO-3: Applying various Data Science techniques to education and social media

CO-4: Analyse various applications of Data Science in health care and bio informatics

COURSE ARTICULATION MATRIX:

(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial)

UNIT-I:

Data Science Applications in various domains, Challenges and opportunities, tools for data scientists. Recommender systems – Introduction, methods, application, challenges.

Principle Component Analysis: For dimensionality and its reduction, Data Visualization with MNIST Dataset, Geometric interpretation of Eigen values, column Standardization.

UNIT-II:

Time Series Data: Stock market index movement forecasting.

Supply Chain Management: Real world case study in logistics.

Support Vector Machines: Geometric interpretation, kernel trick, polynomial kernel, RBF kernel, Domain specific kernel. Decision tree, sample decision tree, building a decision tree with entropy, information gain, Gini-impurity.

UNIT-III:

Data Science in Education, social media.

Text Vectorization: Text2vector, Bag of Words (BOW), Stemming, Unigram, Bigram, TFIDF, word2vector, Avgw2vector, TFIDF Weight edWord2Vector, Count Vectorizing, Ngrams, Implementation of TFIDF, Word2Vector.

UNIT-IV:

Data Science in Healthcare, Bioinformatics,

Introduction to t-SNE, Geometrical interpretation of t-SNE, How to use t-SNE effectively t-SNE with MNIST dataset.

UNIT-V:

Logistic regression, K-nearest neighbour, Naive Bayes algorithm, imbalanced data vs balanced data set.

Case studies in data optimization using Python

TEXT BOOKS:

1. Data Science and its Applications, Aakanksha Sharaff, G. K. Sinha, CRC Press, 2021
2. Data Science: Theory, Analysis and Applications, Q. A. Menon, S. A. Khoja, CRC Press, 2020

REFERENCES:

1. Principal Component Analysis, I. T. Jolliffe, 2nd Edition, Springer
2. Machine Learning, Tom M. Mitchell, McGraw-Hill
3. The Elements of Statistical Learning, Trevor Hastie, Robert Tibshirani, Jerome Friedman, 2nd Edition, Springer

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

B.Tech. Minor (DS) VII Semester

(22MC1DS401) DATA WRANGLING AND VISUALIZATION STRATEGIES

TEACHING SCHEME		
L	T/P	C
3	0	3

EVALUATION SCHEME				
SE	CA	ELA	SEE	TOTAL
30	5	5	60	100

COURSE OBJECTIVES:

- To learn data wrangling techniques
- To introduce and understand the concept of visual perception
- To acquire core skills for visual analysis
- To apply the visualization techniques in real-life applications

COURSE OUTCOMES: After completion of the course, the student should be able to

CO-1: Understand data wrangling and cleaning using NumPy and pandas

CO-2: Apply preprocessing techniques like formatting, outliers, and normalization

CO-3: Create visual representations using various plotting techniques

CO-4: Analyze and interpret multi-dimensional data using visualization techniques

CO-5: Evaluate advanced visualization methods for effective decision-making

COURSE ARTICULATION MATRIX:

(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial)

CO	PROGRAM OUTCOMES (PO)											
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PO-12
CO-1	3	2	2	3	3	2	2	-	2	2	2	3
CO-2	3	2	3	2	3	3	2	-	2	2	2	3
CO-3	3	2	2	3	3	3	1	-	3	2	2	2
CO-4	3	2	2	3	2	2	1	--	3	2	2	3
CO-5	3	2	2	3	2	2	2		3	2	2	3

UNIT-I:

Introduction: Introductory focus on Python Data structure for NumPy & pandas, Auditing data to improve quality.

Data Wrangling: Introduction, Need of data cleanup, data clean up basics.

UNIT-II:

Tasks of Data Wrangling: Data wrangling tools with – formatting, outliers, duplicates, Normalizing and standardizing data. Importance of analytics and visualization in the era of data abundance.

UNIT-III:

Introduction of Visual Perception: visual representation of data, Gestalt principles, information overloads. Creating visual representations, visualization reference model, visual mapping, visual analytics, Design of visualization applications.

Basic Plotting: Line plot - Bar plot - Pie Chart - Scatter Plot - Histogram

UNIT-IV:

Classification of visualization systems, Interaction and visualization techniques misleading, Visualization of one, two and multi-dimensional data, text and text documents, Visualization of groups, trees, graphs, clusters, networks, software, Metaphorical visualization.

UNIT-V:

Visualization of volumetric data, vector fields, processes and simulations, Visualization of maps, geographic information, GIS systems, collaborative visualizations, evaluating visualizations

TEXT BOOKS:

1. Data Wrangling with Python: Tips and Tools to Make Your Life Easier, Jacqueline Kazil and Katharine Jarmul, O'Reilly
2. Interactive Data Visualization: Foundations, Techniques, and Applications, Ward, Grinstein Keim, Natick A. K., Peters Ltd.

REFERENCES:

1. The Visual Display of Quantitative Information, E. Tufte, Graphics Press
2. Data Wrangling with Python: Creating Actionable Data from Raw Sources, Dr. Tirthajyoti Sarkar, Shubhadeep Roy Chowdhury, Packet Publishing, 2019

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

B.Tech. Minor (DS) VII Semester

(22MC2DS401) DATA WRANGLING AND VISUALIZATION STRATEGIES LABORATORY

TEACHING SCHEME		
L	T/P	C
0	3	1.5

EVALUATION SCHEME					
D-D	PE	LR	CP	SEE	TOTAL
10	10	10	10	60	100

COURSE OBJECTIVES:

- To learn data wrangling techniques
- To introduce and understand the concept of visual perception
- To acquire core skills for visual analysis
- To apply the visualization techniques in real-life applications

COURSE OUTCOMES: After completion of the course, the student should be able to

CO-1: Perform experimentation on data wrangling

CO-2: Explain principles of visual perception with datasets

CO-3: Apply visualization techniques for various data analysis tasks

CO-4: Evaluate visualization techniques

CO-5: Apply the visualization in different real time applications.

COURSE ARTICULATION MATRIX:

(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial)

CO	PROGRAM OUTCOMES (PO)											
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PO-12
CO-1	3	2	2	3	3	2	1	-	2	2	2	3
CO-2	3	2	3	2	3	3	2	-	2	2	2	3
CO-3	3	2	2	3	3	3	2	-	3	2	3	2
CO-4	3	2	2	3	2	2	1	-	3	2	2	3
CO-5	3	3	2	3	2	3	2	-	2	2	3	3

LIST OF EXPERIMENTS:

Implement the following experiments using Python

1. Find missing values and perform data imputation
2. Find outliers in a chosen dataset
3. Methods to handle duplicate data
4. Perform data normalization
5. Explore 2-D charts such as Clustered bar charts, connected dot plots, pictograms, bubble charts, radar charts, polar charts, Range chart, Box-and-whisker plots, univariate scatter plots, histograms word cloud, pie chart, waffle chart, stacked bar chart, tree map.
6. Multi-dimensional data visualization
7. Graph data visualization

TEXT BOOKS:

1. Data Wrangling with Python: Tips and Tools to Make Your Life Easier, Jacqueline Kazil and Katharine Jarmul, O'Reilly
2. Data Visualization A Handbook for Data Driven Design, Andy Kirk, Sage Publications, 2016
3. Understanding Data with Graphs, Philipp K. Janert, Gnuplot in Action, Manning Publications, 2010

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

B.Tech. Minor (DS) VIII Semester

(22MC1DS402) TIME SERIES ANALYSIS AND FORECASTING

TEACHING SCHEME		
L	T/P	C
3	0	3

EVALUATION SCHEME				
SE	CA	ELA	SEE	TOTAL
30	5	5	60	100

COURSE OBJECTIVES:

- To learn basic analysis of time series data and time series regression
- To learn auto-regressive and model averaging models
- To learn basic concepts of forecasts for a time series data
- To learn how to perform the advanced computation like Multivariate Time Series Models and Forecasting

COURSE OUTCOMES: After completion of the course, the student should be able to

CO-1: Apply ideas to real time series data and interpret outcomes of analyses.

CO-2: Learn how to construct a time-series plot & identifying the underlying patterns in the data

CO-3: Learn basic concepts featurization techniques in time series analysis

CO-4: Learn how to develop forecasts for a time series that has a seasonal pattern

CO-5: Demonstrate an advanced understanding of the underlying concepts in the time series and frequency domains for computation, visualization, and analysis of time series data

COURSE ARTICULATION MATRIX:

(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial)

CO	PROGRAM OUTCOMES (PO)											
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PO-12
CO-1	3	3	2	3	3	2	-	-	2	2	2	3
CO-2	3	2	3	2	3	3	-	-	2	2	2	3
CO-3	3	2	3	3	3	3	-	-	3	2	3	2
CO-4	3	2	2	3	2	2	-	-	3	2	2	3
CO-5	3	2	2	3	2	3	-	-	2	2	3	2

UNIT-I:

Introduction of time series analysis, The Nature of Time Series Data, Internal structures of time series, Time Series Statistical Models, Moving Window for Time series data.

UNIT-II:

Fourier decomposition, deep learning features LSTM, Image Histogram, Deep learning feature CNN, Relational data, Graph data, Indicator Variable, Feature binning, interaction variable.

UNIT-III:

Mathematical transforms, Model specific featurizations, Feature orthogonality, Domain specific featurizations, Feature slicing.

Kaggle Winner Solutions: Students are suggested to carry out with at least 3 Kaggle winnercompetition case study solutions and present the report for that. (<https://www.kaggle.com/code/sudalairajkumar/winning-solutions-of-kaggle-competitions>)

UNIT-IV:

Autocorrelation and Partial autocorrelation, General Approach to Time Series Modelling and Forecasting, Evaluating and Monitoring Forecasting Model Performance, Introduction Least Squares Estimation in Linear Regression Models, Statistical Inference in Linear Regression.

UNIT-V:

Model Adequacy Checking, Regression Models for General Time Series Data. Exponential Smoothing, First order and Second order, Multivariate Time Series Models and Forecasting, Multivariate Stationary Process, ARIMA model, Vector AR (VAR) Models.

TEXT BOOKS:

1. Introduction to Time Series Analysis and Forecasting, Douglas C. Montgomery, Cheryl L. Jen, 2nd Edition, Wiley Series in Probability and Statistics, 2015
2. Master Time Series Data Processing, Visualization, and Modeling Using Python, Dr. Avishek Pal, Dr. P. K. S. Prakash, 2017

REFERENCES:

1. Introduction to Time Series Analysis, P. J. Brockwell and R. A. Davis
2. Time Series Analysis and Its Applications, Robert H. Shumway and David S. Stoffer
3. Introduction to Statistical Time Series, W. A. Fuller
4. Time Series Analysis, Wilfredo Palma